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Research Paper

A spreadsheet-based technique to calculate the passive soil pressure based on the log-spiral method

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ABSTRACT

This paper presents an efficient technique to calculate the passive earth pressure of a cohesionless sloping backfill on a rigid retaining wall which is an extension of the Terzaghi's log-spiral approach. Moment equilibrium conditions are used to compute the minimum value of the passive pressure required to mobilize the passive failure. Terzaghi assumed that the pole of the log-spiral failure surface lies inside the soil mass, whereas the proposed technique and solutions show that the pole can lie inside or outside the soil mass. Thus, the developed technique has extended the log-spiral method reported by Terzaghi by providing a generalised solution. The distinctive feature of the proposed technique is the explicit and rigorous solutions developed which allow to evaluate the passive earth pressure with different friction angles for the soil and wall-soil interface, backfill slopes and wall inclination angles. These solutions are relatively simple to incorporate into a spreadsheet to derive the passive force and pressure. A series of experiments were carried out using a physical model and the Particle Image Velocimetry (PIV) technique to analyse the failure surface. The developed solutions were then validated with data obtained from model tests as well as experimental data found in the literature.

1. Introduction

The lateral earth pressure is one of the primary parameters influencing the interaction between the soil and adjacent structural members. Therefore, estimation of the earth pressure is vital when the design of such structural members is sought. While many retaining structures are designed considering an active or at-rest state for the backfill, in practice there are certain cases, where the interaction mobilizes a passive earth pressure. Such cases require adequate estimation of the passive earth pressure.

Classical theories, such as Coulomb (1973) and Rankine (1857) are widely used as the fundamental methodologies to estimate the lateral passive pressure of the backfill on retaining walls. For instance, most of the American states use Rankine triangular earth pressure in the estimation of passive earth pressure behind the abutments of Integral bridges (Kunin and Alampalli, 2000). They use the limit equilibrium approach to calculate the passive earth pressure and the coefficient of passive pressure, k_p. Although the cited theories involve a number of differences, they all assumed the passive failure surface to be planar. Coulomb (1973), assumed that the failure occurs along a plane surface that passes through the toe of the wall at an angle θ with the horizontal such that the passive pressure acting on the retained structure is a minimum. Rankine (1857), on the other hand stipulated that the passive failure surface is a plane, inclined at $(45 - \varphi/2)$ to the principle horizontal axis (where φ represents the internal friction angle of the soil). However, the assumption of a planar failure surface can be problematic when the wall surface is not smooth. According to Terzaghi (1943) and Terzaghi et al. (1996), the failure surface is not a plane when the angle of the interface friction δ between the wall and the retained backfill is more than zero. Terzaghi et al. (1996) stated, when the angle δ is greater than $\varphi/3$ the curvature of the failure surface becomes significant and hence this curvature must be taken into consideration in the calculation of the passive earth pressure.

In a real practical case, the interface friction between the wall and the soil is far from being ideally smooth, hence appropriate estimation of the passive earth pressure must account for the curvature in the failure surface. Terzaghi (1943) stated that sufficiently accurate results can be obtained when a logarithmic spiral arc is used to represent the bottom part of the rupture surface (curve *bg* in Fig. 1) while the rest of the surface is a tangential line to the log-spiral curve rising up at an angle

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equal to $(45-\varphi/2)$ with the horizontal.

The log-spiral method proposed by Terzaghi (1943) was used by a number of experimental studies and showed to be in closer agreement, with the measured data, than Rankine or Coulomb methods. Duncan and Mokwa (2001) studied the passive earth pressure mobilized due to the translation of 1 m high concrete bulkhead embedded in the soil and concluded that the log-spiral method is the most accurate theory for estimating the passive earth pressure over a wide range of interface friction angles. Fang et al. (2002) reported experimental results from testing using dry sand under monotonic rigid wall movements. They stated that the log-spiral method proposed by Terzaghi can successfully replicate the passive earth pressure by using the critical state concept. Wilson and Elgamal (2010) conducted large scale model tests using granular soil to measure the passive earth pressure. They reported that the log-spiral prediction provided very good estimation by comparing with the measured data. Rollins et al. (2017) reported that the lateral passive pressure calculated using the log-spiral method replicates the measured force within an error of less than 10% compared to 47% and 60% respectively, using Coulomb and Rankine methods.

Although the log-spiral theory can provide an accurate means to estimate the passive earth pressure, design engineers often prefer Rankine and Coulomb methods, due to their simplicity, despite of the potential misrepresentation of the actual conditions. The primary reason behind that is the lack of a rigorous solution to evaluate the passive earth pressure using the log-spiral theory. Even though, the latter was developed a few decades ago, (Terzaghi 1943), limited research studies utilized the log-spiral composite failure to estimate the static passive earth pressure (Terzaghi, 1943; Terzaghi et al., 1996; Patki et al., 2015; Liu et al., 2018), and none have developed comprehensive easy-to-implement solutions. Terzaghi's (1943, 1996) solution is proposed only for the cases when the pole of the log-spiral is located inside the slip mass; Liu et al. (2018) proposed a complicated solution based on the pole outside the slip mass. The solution developed by Patki et al. (2015) does not apply to sloping backfills.

This study aims to develop a comprehensive and rigorous numerical solution, based on the log-spiral theory, such that it can be easily formulated on a spreadsheet to estimate the passive pressure of the soil acting on a rigid retaining wall. The proposed solution is versatile and can examine different internal friction angles, wall-soil interface friction angles, wall inclinations, backfill slopes and external surcharge, and covering the possibility of the pole located either inside or outside the slip mass. In this paper, the proposed solution will be referred to as the "log-spiral solution".

2. Properties of the failure surface

Fig. 1 shows a hypothetical case, where a rigid wall retains a homogenous deposit of dry cohesionless soil with a unit weight of $\gamma$and friction angle of φ. The backfill is inclined at an angle β with the horizontal while the wall face is inclined at an angle ω to the vertical. Imagine the wall translates monotonically against the backfill until a localized plastic equilibrium is reached in the soil close to the wall. This state is expressed by the mobilization of a passive failure in the retained soil. The failure occurs when the passive wedge *abge* is at the verge of sliding along *bge*. Therefore, the latter is often termed as the sliding or the failure surface.

The composite failure surface *bge*, in the case prescribed in Fig. 1, is based on the hypotheses of Terzaghi (1943). Hence, it comprises a logarithmic spiral arc, *bg*, and a straight line *ge*, which is a tangent to the spiral at point *g*. The passive wedge consists of the area *abg* (outlined by the log-spiral arc) and the Rankine passive zone outlined by the isosceles triangle *age* (Terzaghi et al. 1996). Liu et al. (2018) stated that the angles α_1 and $\alpha_2$for a cohesionless sloped backfill can be calculated using the following relationships,

$$\alpha_1 = \frac{\pi}{4} - \frac{\varphi}{2} + \frac{1}{2} arcsin\left(\frac{sin\beta}{sin\varphi}\right) - \frac{\beta}{2} \tag{1}$$

$$\alpha_2 = \frac{\pi}{4} - \frac{\varphi}{2} - \frac{1}{2} arcsin\left(\frac{sin\beta}{sin\varphi}\right) + \frac{\beta}{2} \tag{2}$$

Therefore, when the backfill is horizontal ($\beta = 0$), $\alpha_1 = \alpha_2 = \frac{\pi}{4} - \frac{\phi}{2}$.

The force P_p that mobilises the passive failure acts on the wall at an angle $(\delta - \omega)$ with the horizontal, whereas δrepresents the interface friction angle.

The log-spiral arc, *bg*, is given by the formula,

$$r_i = r_o e^{\theta_i tan\varphi} \tag{3}$$

where r_o is the initial radius of the spiral arc at point *b*, r_i represents the radius of the spiral at any random point *i* on the spiral arc, $\theta_i$is the angle of rotation subtended by r_o and r_i, and φ represents the internal friction angle of the soil. Hence, at point *g*, the Eq. (3) can be written as,

$$r_g = r_o e^{\theta_g tan\varphi} \tag{4}$$

It is apparent that the size of the spiral arc is characterized by r_o, r_g, θ_g and φ. While φ is taken as a constant for a given type of soil, the remaining parameters are functions of the virtual location of the Pole O of the log-spiral. The location of the Pole O, therefore, is essential in developing a generalised solution for the problem illustrated in Fig. 1.

Logically, the Pole O can be located either outside the soil mass

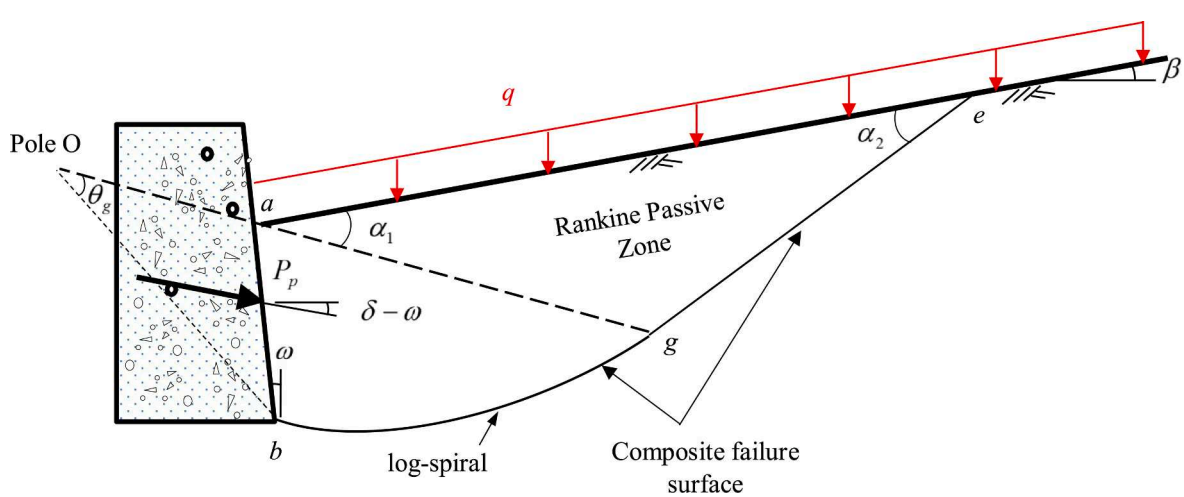

Fig. 1. The composite log-spiral failure surface.

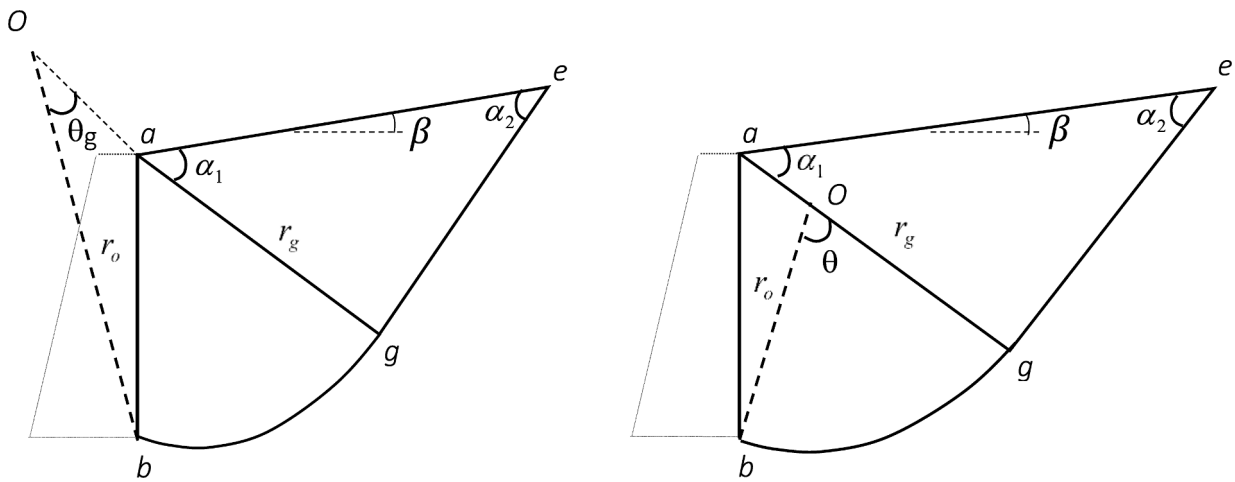

Fig. 2. The failure surface when the spiral Pole is (a) outside the soil mass and (b) inside the soil mass.

(Fig. 2a) or inside the soil mass (Fig. 2b). While the Pole can also fall at point *a*, this can be treated as a unique limiting case of either (a) or (b) above (the Pole lies outside and inside the soil mass, respectively) but not as an independent case. Terzaghi et al. (1996) suggested that the Pole lies on the line O*g* inside the soil mass (i.e., case b above). However, they did not discuss the possibility of the Pole to be located outside the soil mass. Until recently, Liu et al. (2018) developed a mathematical formulation to estimate the passive earth pressure coefficient, k_p, assuming that the Pole of the spiral falls outside the soil mass. However, developing a solution based on the assumption of the Pole either inside or outside the soil mass only may not cover all potential solutions. Moreover, the solutions developed by Liu et al. (2018) are based on minimising the passive force objective function expressed in terms of the "design variable" which is the y-coordinate of an intersection point vertically above point *g* and horizontally aligned with point *a*, in Fig. 2. As will be shown later, this paper proposes an alternative, simpler "design variable" ξ which is based on the distance O*a*. Patki et al. (2015) proposed a numerical solution to determine the lateral earth pressure using the log-spiral method assuming an arbitrary location of the Pole O along the spiral radius. However, their solution is limited to horizontal backfill ($\beta = 0$) and does not apply for sloping backfill. Therefore, the present study aims to develop a generalised solution that examines both cases of Pole O location (inside and outside the soil mass) and applies to horizontal and sloping backfill, vertical and inclined wall and to a wide range of interface friction angles.

From the simplified description above, it is understood that the magnitude of the force due to passive soil pressure, P_p, in Fig. 1, can be calculated using the principle of the limit equilibrium. This requires accounting for the forces acting on the passive wedge, the wall and the failure surface. These forces include the self-weight of the passive soil wedge, the soil reaction forces acting along the slip surface in addition to the force P_p. In order to achieve a determinate solution for the problem, the area *fge* of the passive wedge is replaced by a Rankine passive pressure triangle as shown in Fig. 3 (Terzaghi et al. 1996). However, the system given in Fig. 3 is just an illustration of the free body diagram of the failure wedge with an arbitrary location of the Pole O outside the soil

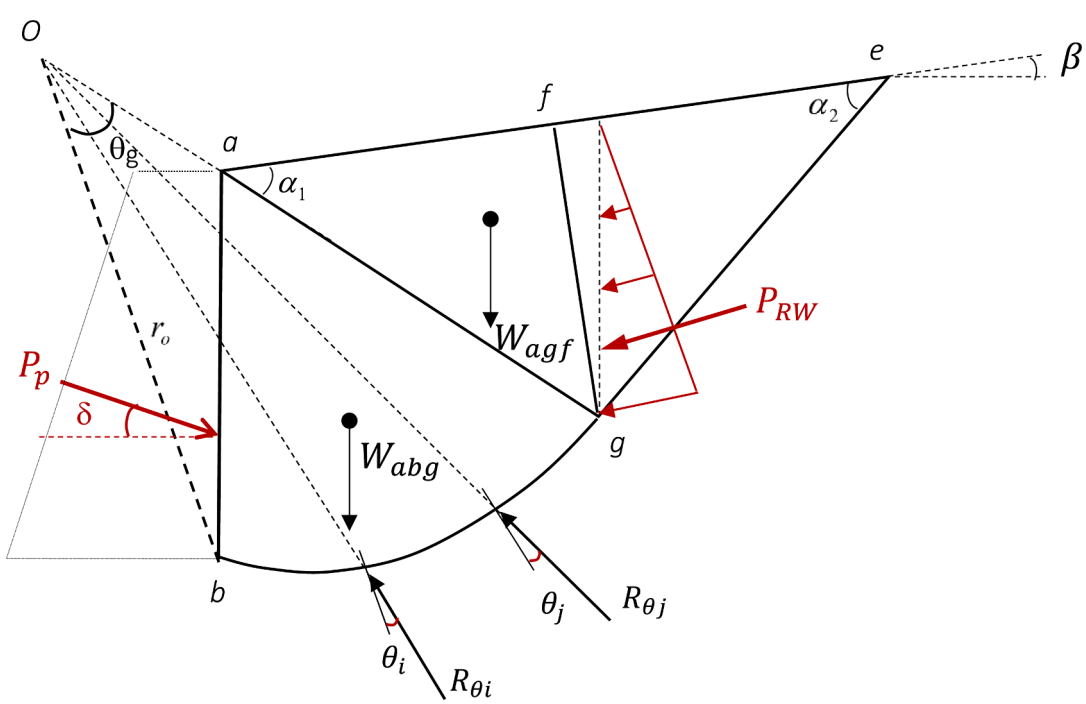

Fig. 3. Free body diagram of the failure wedge.

mass (Case-a). As the location of the Pole is normally unknown, it is impossible to figure out the characteristics of the failure surface and accordingly the passive earth pressure. A meaningful approach to deal with such a problem is to find the unique solution given as the minimum magnitude of the passive earth pressure corresponding to the failure surfaces defined by the parameter ξ.

3. Formulation of the limiting equilibrium state

As described earlier, the proposed log-spiral solution is formulated based on the principles of the limit equilibrium. Hence, the moment equilibrium about the Pole O is applied. This however, depends heavily on the location of the Pole inside or outside the soil mass. Therefore, the solution will be presented firstly for Case-a (Pole outside the soil mass) and then for Case-b (Pole inside the soil mass) separately.

3.1. Case-a: the pole lies outside slip mass

Let us assume that the Pole O lies on the line O*a* outside the slip mass *abge* (see Fig. 4 below) at an arbitrary distance ξ ("design parameter") from point *a*. Pole O slides along the direction of O*a* to generate a range of log-spiral curves *bg*. According to the given assumptions, the following relationships can be established,

$$\text{H(the vertical projection of the retained length ab)} = |ab|cos\omega \tag{5}$$

$$x'\,(\textit{the horizontal offset between points a and b}) = |ab|sin\omega \tag{6}$$

$$\eta = \alpha_1 - \beta \tag{7}$$

$$\text{Distance } Oa = |\xi| \tag{8}$$

In a given trial, let ξ be specified (note: ξ is taken as negative when the Pole lies outside the soil mass as in Case-a), and based on that, the following parameters can be computed:

Coordinates of point *a* (x_a, y_a):

$$x_a = |\xi|cos\eta \tag{9}$$

$$y_a = |\xi|sin\eta \tag{10}$$

$$r_o = \sqrt{(H + y_a)^2 + (x_a + x')^2} \tag{11}$$

Coordinates of point $b(x_b,\ y_b)$:

$$x_b = x_a + |x'| \tag{12}$$

$$y_b = y_a + H \tag{13}$$

$$\theta_g = sin^{-1}\frac{y_b}{r_o} - \eta \tag{14}$$

Coordinates of point *g* (x_g, y_g):

$$x_g = r_g cos\eta \tag{15}$$

$$y_g = r_g sin\eta \tag{16}$$

$$|ag| = r_g - |\xi| \tag{17}$$

$$|fg| = |ag|sin\alpha_1 \tag{18}$$

Coordinates of point *f* (x_f, y_f):

$$x_f = x_g - |fg|sin\beta \tag{19}$$

$$y_f = y_g - |fg|cos\beta \tag{20}$$

The slip mass *abge* is also defined, including the log-spiral *bg*, the coordinates of which can be established using the log-spiral relationship (equation (3)), where θ is the angle subtended between the log-spiral radius *r* and r_o (see Fig. 4). The coordinates of point *p* on the log-spiral curve are then given by:

$$x_p = rsin(\theta + \lambda) \tag{21}$$

$$y_p = rcos(\theta + \lambda) \tag{22}$$

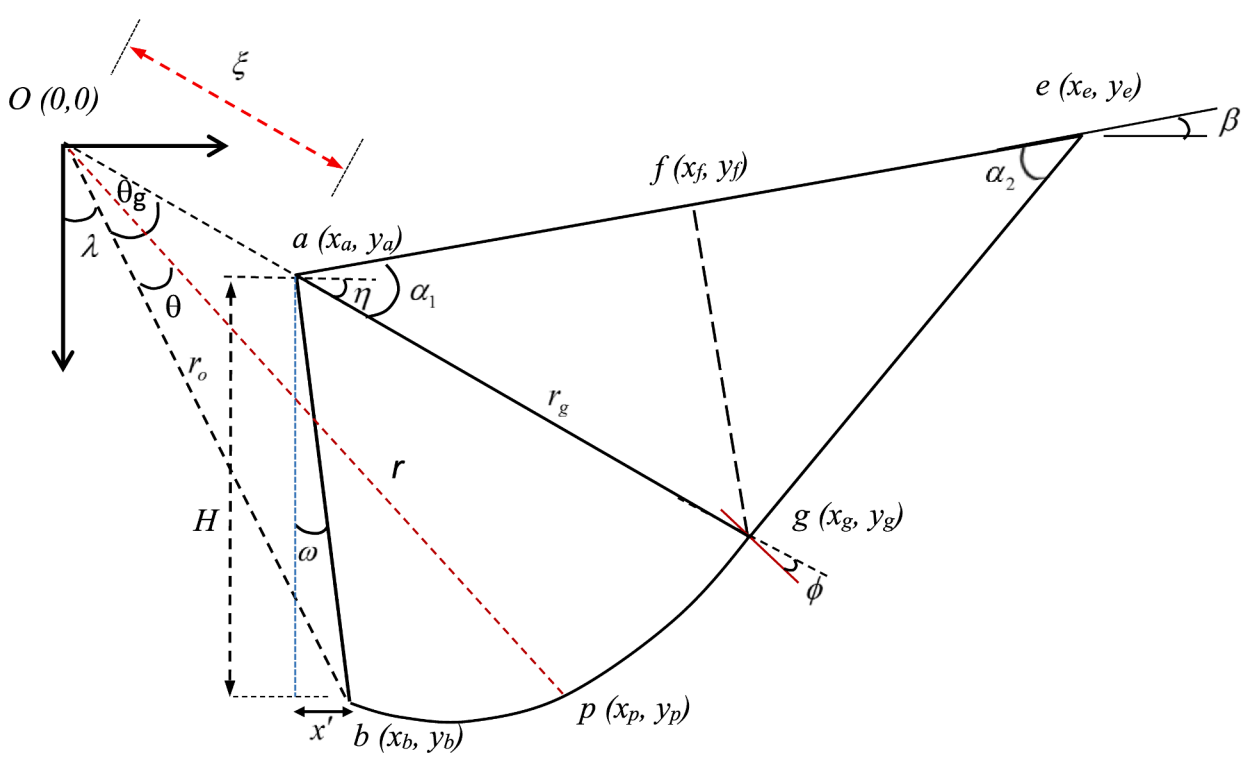

Fig. 4. Slip mass when the Pole of the spiral is outside the slip mass.

where λ is an angle subtended by r_o and the vertical axis.

The area moment of *abg* about the Pole O is given by,

$$M^a_{abg} = M^a_{obg} - M^a_{oba} \tag{23}$$

where M^a_{obg} is the area moment of the log-spiral sector *obg* and M^a_{oba} is the area moment of the triangle *oba* (see Fig. 4). Now an infinitesimal sector with subtended angle $d\theta$ of the log-spiral is considered in Fig. 5. The centroid of that sector is located at $\frac{2}{3}r$ from the Pole O.

Then the anticlockwise incremental area moment dM^a about O is given by:

$$dM^a = \frac{1}{2}d\theta r^2 . \frac{2}{3} r sin(\theta + \lambda) \tag{24}$$

Expanding and considering that $r = r_o e^{\theta tan\varphi}$ gives:

$$dM^a = \frac{1}{3} r_o^3 e^{3\theta tan\varphi} (sin\theta cos\lambda + cos\theta sin\lambda) d\theta \tag{25}$$

which leads to,

$$M^a_{obg} = \frac{1}{3} r_o^3 \int_0^{\theta_g} e^{3\theta tan\varphi} (sin\theta cos\lambda + cos\theta sin\lambda) d\theta \tag{26}$$

Integrating by parts, Eq. (27) is obtained:

$$M^a_{obg} = \frac{r_o^3 cos\lambda}{3(1 + 9tan^2\phi)} \left[1 + e^{3\theta_g tan\phi}\left(3tan\phi sin\theta_g - cos\theta_g\right)\right] + \frac{r_o^3 sin\lambda}{3(1 + 9tan^2\phi)} \left[e^{3\theta_g tan\phi}\left(sin\theta_g + 3tan\phi cos\theta_g\right) - 3tan\phi\right] \tag{27}$$

Similarly, the anticlockwise moment arm of triangle *oba* is,

$$\bar{x}_{oba} = \frac{1}{3}(x_a + x_b) \tag{28}$$

so that,

$$M^a_{oba} = \frac{1}{6}(x_a y_b - x_b y_a)(x_a + x_b) \tag{29}$$

The anticlockwise moment due to self-weight of slip mass abg about O is:

$$M_{abg} = \left(M^a_{obg} - M^a_{oba}\right)\gamma \tag{30}$$

Considering Fig. 6, the following relationships can be derived,

$$\upsilon = \eta - \delta + \omega \tag{31}$$

$$|ka| = |\xi| sin\upsilon \tag{32}$$

$$|da| = \left(\frac{2}{3}|ab|\right) cos\delta \tag{33}$$

$$L_1 = |ka| + |da| \tag{34}$$

$$L_{RW} = r_g sin\alpha_1 - \left(\frac{1}{3}|fg|\right) \tag{35}$$

$$\text{Surcharge load, } V = q|af'| \tag{36}$$

$$\text{where } af' = |af| + |fg| tan\beta \tag{37}$$

$$af = \frac{|x_f - x_a|}{cos\beta} \tag{38}$$

M_q is the moment produced by the surcharge q about the Pole O

$$M_q = V\left|x_a + \left(\frac{x_{f'} - x_a}{2}\right)\right|, \text{ where } x_{f'} = x_g \tag{39}$$

$$P_{RW} = \frac{1}{2} k_p \gamma H^2_{RW} cos\beta \tag{40}$$

where H_{RW} is calculated as,

$$H_{RW} = \frac{|fg|}{cos\beta} \tag{41}$$

k_p is Rankine's passive pressure coefficient for sloped backfill and calculated as,

$$k_p = \frac{cos\beta + \sqrt{cos^2\beta - cos^2\varphi}}{cos\beta - \sqrt{cos^2\beta - cos^2\varphi}} \tag{42}$$

$$M_{agf'} = \frac{1}{6}|af'||fg|\gamma\left(x_a + x_g + x_{f'}\right) \tag{43}$$

$$F_q = qcos\beta k_p H_{RW} \tag{44}$$

$$L_q = r_g sin\alpha_1 - \left(\frac{|fg|}{2}\right) \tag{45}$$

$$M_{qh} = F_q L_q \tag{46}$$

By taking moment equilibrium about O:

$$P_p = \frac{1}{L_1}\left(P_{RW} L_{RW} + M_{qh} + M_{abg} + M_q + M_{agf'}\right) \tag{47}$$

Finally, passive force = min P_p.

For a given wall geometry (wall height and slope angles) and set of geotechnical properties (internal friction angle, unit weight of soil and wall-soil interface friction), the passive force P_p is a function of the ξ through the parameters L_1, P_{RW}, L_{RW} etc. in equation (47). Thus, a spreadsheet can be easily set up to compute P_p and determine its minimum value as a function of ξ.

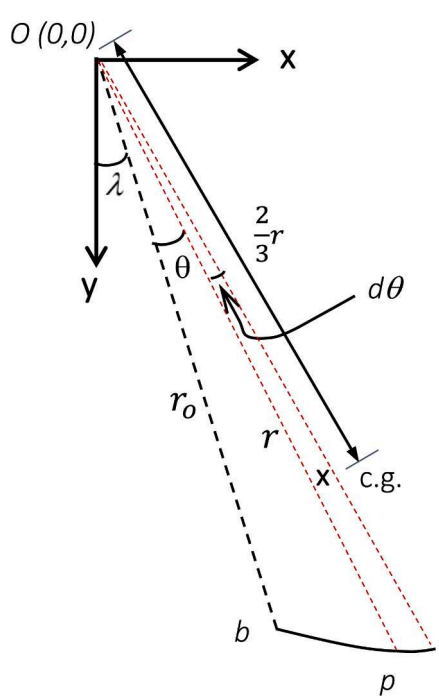

Fig. 5. Infinitesimal sector of log-spiral in Case-a.

3.2. Case-b: the pole lies inside slip mass

In Case-b, the Pole is assumed to be located inside the slip mass as shown in Fig. 7. Under this assumption, the area moments are different

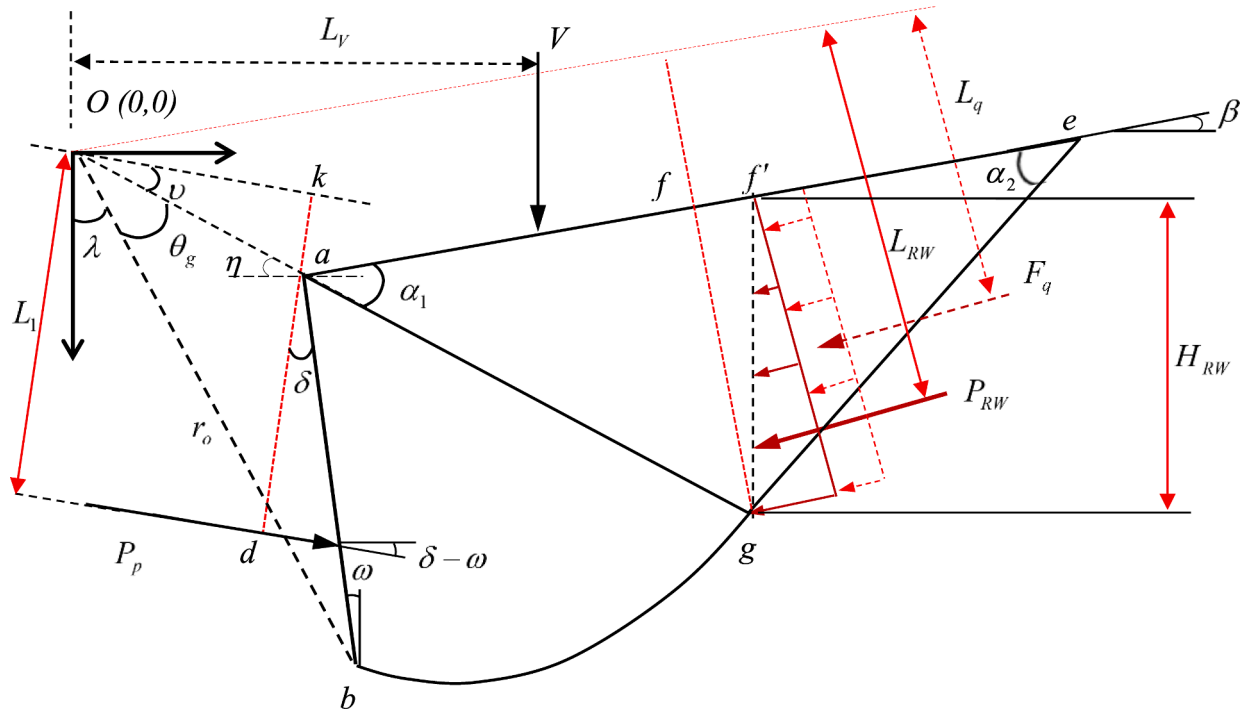

Fig. 6. Forces and moment arms about the Pole O in Case-a.

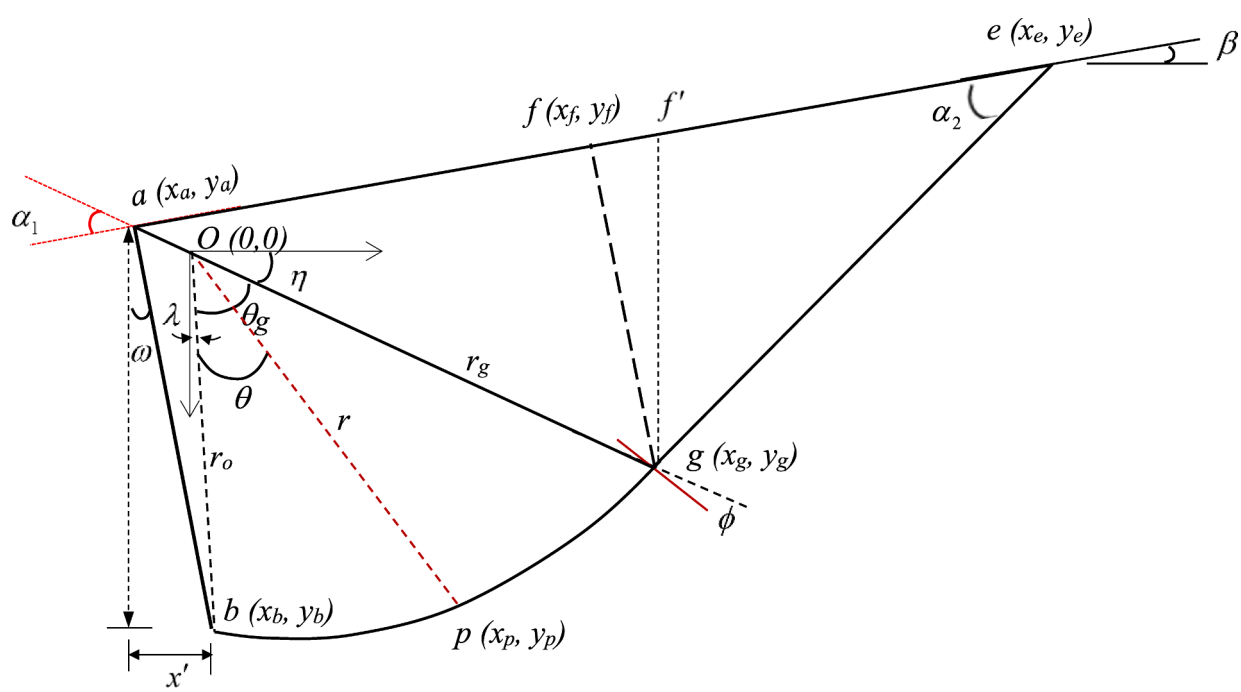

Fig. 7. Slip mass when the Pole of the log-spiral is inside the slip mass.

compared to those discussed for Case-a.

Similar to Case-a, at a given ξ (note: ξ is taken as positive when the Pole lies inside the soil mass) the following relationships can be established,

Coordinates of point a,

$$x_a = -(\xi cos\eta) \tag{48}$$

$$y_a = -(\xi sin\eta) \tag{49}$$

Coordinates of point b,

$$x_b = x' + x_a \tag{50}$$

$$y_b = H + y_a \tag{51}$$

$$r_0 = \sqrt{x_b^2 + y_b^2} \tag{52}$$

$$\lambda = sin^{-1}\frac{x_b}{r_o} \tag{53}$$

$$\theta_g = 90^\circ - \lambda - \eta \tag{54}$$

$$r_g = Og = r_o e^{\theta_g tan\varphi} \tag{55}$$

Coordinates of point g,

$$x_g = r_g cos\eta \tag{56}$$

$$y_g = r_g sin\eta \tag{57}$$

$$ag = |\xi| + r_g \tag{58}$$

$$fg = |ag| sin\alpha_1 \tag{59}$$

Coordinates of point f,

$$x_f = x_g - (fg sin\beta) \tag{60}$$

$$y_f = y_g - fg cos\beta \tag{61}$$

$$x_p = r sin(\theta + \lambda) \tag{62}$$

$$y_p = r cos(\theta + \lambda) \tag{63}$$

Calculation of the area moments of oba, agf' (see Fig. 8)

$$M^a_{oba} = \frac{1}{6}(x_a + x_b)(x_a y_b - x_b y_a) \tag{64}$$

$$M^a_{agf'} = \frac{1}{6}|af'||fg|\gamma(x_a + x_g + x_{f'}), \text{ where } x_{f'} = x_g \tag{65}$$

The area moment of obg is calculated in the same way for Case-a, equation (27). Accordingly, the moment of the wedge $abgf'$ is given by,

$$M_{abgf'} = \left(M^a_{obg} + M^a_{oba} + M^a_{agf'}\right)\gamma \tag{66}$$

Now, by taking moment equilibrium about the Pole O for the forces shown in Fig. 8,

$$L_1 = |at| - |ak| \tag{67}$$

$$|at| = \frac{2}{3}|ab| cos\delta \tag{68}$$

$$|ak| = |\xi| sin\upsilon \tag{69}$$

where $\upsilon = \eta - \delta + \omega$.

Calculation of the passive zone forces $P_{RW}$and F_q are shown below,

$$H_{RW} = \frac{|fg|}{cos\beta} \tag{70}$$

$$P_{RW} = \frac{1}{2}\gamma H^2_{RW} k_p cos\beta \tag{71}$$

$$F_q = q cos\beta k_p H_{RW} \tag{72}$$

$$L_{RW} = \frac{2}{3}|fg| - |\xi| sin\alpha_1 \tag{73}$$

$$af = |ag| cos\alpha_1 \tag{74}$$

$$V = q|af'|,$$

where

$$af' = af + fg tan\beta \tag{75}$$

$$L_V = \left|\left(\frac{(|x_a| + |x_{f'}|)}{2}\right) - |x_a|\right| \tag{76}$$

$$M_{qh} = F_q L_q \tag{77}$$

$$L_q = \frac{1}{2}|fg| - |\xi| sin\alpha_1 \tag{78}$$

Finally,

$$P_p = \frac{1}{L_1}\left(P_{RW} L_{RW} + V L_V + M_{qh} + M_{abgf'}\right) \tag{79}$$

Passive force = min P_p.

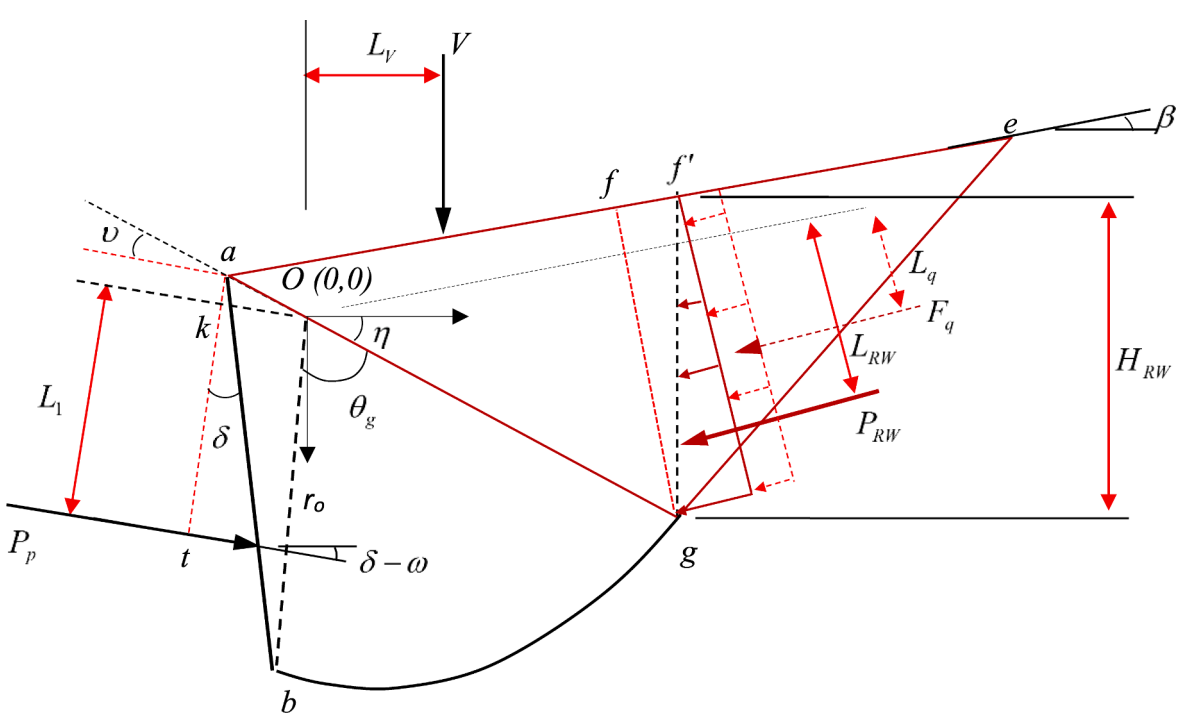

Fig. 8. Forces and moment arms about Pole O in Case-b.

4. Validation of the solution

In order to validate the solution discussed above, a series of physical model tests were carried out on a loose dry sand. The tests were conducted using the test chamber shown in Fig. 9 below. The chamber includes a 700 mm × 250 mm × 300 mm tank to accommodate the soil and a 250 mm × 300 mm × 13 mm steel wall to retain the soil. The front side of the tank is made of a 50 mm thick clear acrylic panel to allow for visual observation. The steel wall is supported by two metal bars, to maintain pure translational movement, and being moved using a threaded movement bar. A load cell is placed between the movement bar and the wall to measure the total applied force. The movement is monitored during the tests using a displacement dial gauge.

The soil used in the experiments is a dry uniform sand with a grain size ranging between 150 and 300 μm. The unit weight of the sand used in each test was calculated by measuring the weight of the sand and the volume occupied. Direct shear tests were conducted to determine the internal friction angle of the sand as well as the sand-wall and sand-acrylic interface friction angles. Table 1 summarises the properties of the sand used in the tests.

Displacement-controlled passive wall movement was applied by spinning the movement bar clockwise, which pushes the wall against the soil. The fine threads in the movement bar helped to achieve a slow and uniform displacement throughout the test.

4.1. Experimental results

The primary objective of the laboratory experiments in this study is to obtain real data about the passive earth pressure. Therefore, attention was given to estimate the frictional forces between the wall and the sides/base of the tank as well as between the sand and the sides of the tank. Subsequently these forces were subtracted from the total measured force. Fig. 10 shows the passive force–displacement curves for three different experiments. In these experiments the height of the retained soil varied to obtain multiple validation data points.

The maximum passive forces for the three tests in Fig. 10 are summarized in Table 2. The log-spiral solution developed in this study was formulated in a spreadsheet, and then used to calculate the passive earth pressures for the same conditions prescribed in the experiments. The passive force of each test is determined as the minimum force in the $P_p - \xi$ diagram. Fig. 11 shows the $P_p - \xi$ plots for the three tests.

The positive values of the distance ξ in Fig. 11 refer to the Pole of the log-spiral located inside the soil mass, while the negative values refer to the Pole located outside the soil mass. It is evident that the Pole in all of the three tests was located outside the soil mass but at different radial distances.

The log-spiral results are then compared with the corresponding experimental results as well as those calculated using Coulomb and Rankine methods. In this case, the magnitudes of the angles β, ω and the surcharge q are equal to zero.

Table 1
Properties of the soil used in the tests.

Unit weight (kN/m^3)	15.3
Internal friction angle	35.5°
Sand-wall friction angle	24°
Sand-acrylic friction angle	25°

Table 2 illustrates a comparison between the measured and calculated data. It is apparent that the passive forces calculated using the developed log-spiral solution correlate well with the measured data. The passive forces determined using the Coulomb method overestimates the measured forces, while the Rankine method significantly underestimates the passive forces.

In an attempt to verify the assumption about the shape of the failure surface, which consists of a curved part (presumably spiral arc) and a tangent line, the actual failure surface was investigated using images captured during the experiments. The PIV technique developed by Stanier et al. (2015) was used to capture the actual failure surface developed in one of the experiments (retained height = 0.24 m). The failure surface can be clearly observed in Fig. 12 (Sigdel et al., 2021).

Fig. 12a shows the contour map for total displacements plotted across the soil mass. The failure surface can be seen along the dotted line, which consists of a curved part at the bottom and a tangent line rising towards the soil surface. Similarly, the shear band plot in Fig. 12b clearly indicates the failure surface, which consists of a curved part and a straight line.

In order to further confirm the validation of the log-spiral solution proposed in this paper against real data measured at a larger scale, reference was also made to experimental results reported by Fang et al. (2002). They conducted a series of tests on dry Ottawa sand to measure the passive pressure acting on a rigid steel wall under a monotonic translational movement. The properties of the backfill used by Fang et al. (2002) are summarized in Table 3.

The proposed log-spiral solution was used to calculate the passive earth pressure based on the critical state properties. The results show that the proposed method successfully replicated the measured data within an error of less than 10% as shown in Table 4. The log-spiral solution has also been used to produce results for the coefficient of passive earth pressure $k_p$and compared with those reported by Shiau et al. (2008) and other researchers. They reported results of $k_p$calculated using different methods. The comparison is shown in Table 5. It is apparent that the results of the present log-spiral solution fits well in the admissible solution field of Shiau et al. (2008). They also closely agree with the data reported by Caquot and Kerisel (1948).

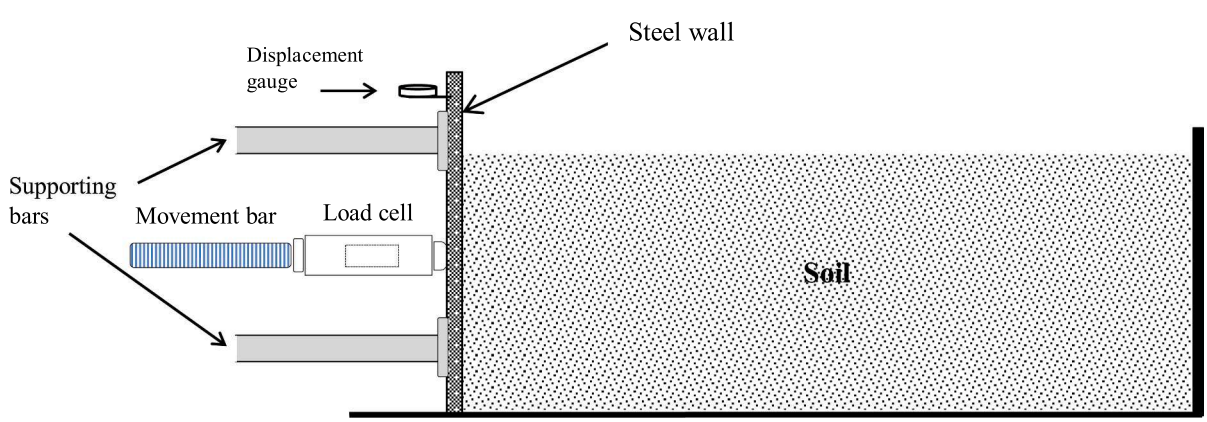

Fig. 9. The test chamber.

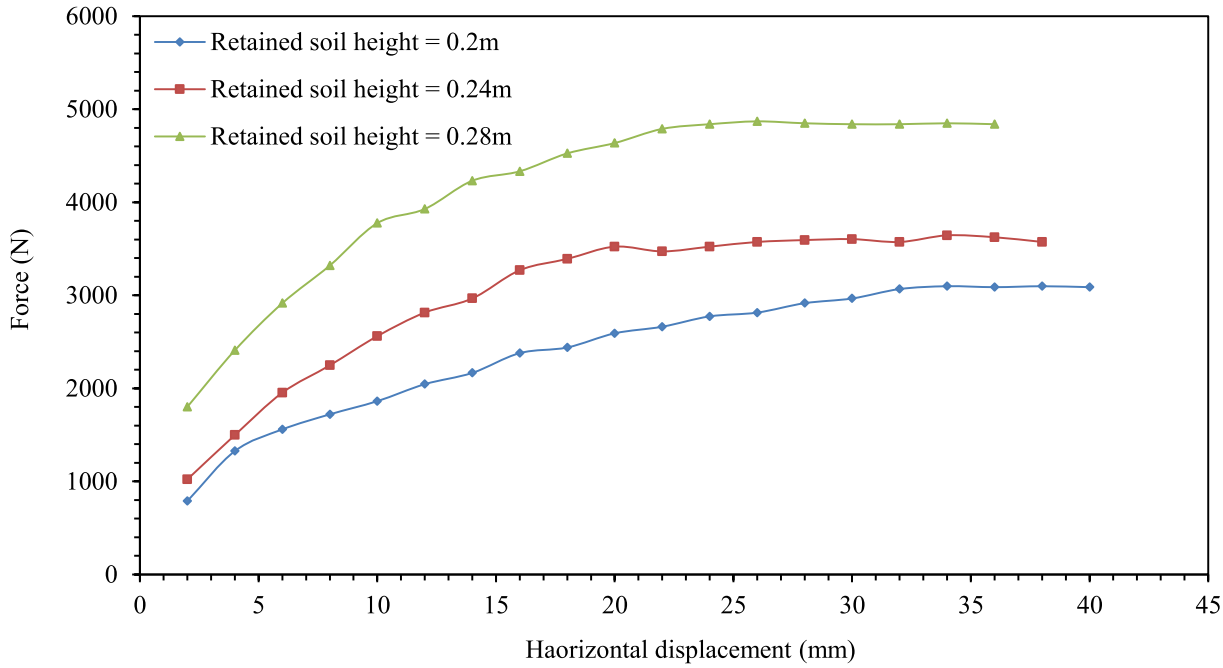

Fig. 10. Passive force vs Horizontal displacement.

Table 2
Comparison between the calculated and measured passive forces.

Test	Retained height (m)	Passive soil force (N)				
		Measured	Log-spiral solution (this paper)		Coulomb	Rankine
			Force	ξ(m)		
1	0.2	3067.7	2511	−0.16	3286	1153.8
2	0.24	3593.6	3614.4	−0.2	4732.6	1661.4
3	0.28	4837	4919.1	−0.24	6441.6	2261.4

4.2. Effects of the wall-soil interface friction

According to Terzaghi et al. (1996), the angle of friction between the backfill and the retaining structure influences the curvature of the log-spiral arc and consequently on the passive soil pressure acting on the wall. Hence, it is worthwhile to investigate the interaction behaviour at different interface friction angles. The log-spiral solution developed in this study was utilized as an efficient and simple tool to conduct this investigation. In order to investigate the influence of the wall-soil interface friction, the magnitudes of angles β, ω and the surcharge q are equated to zero.

Fig. 13 shows the relationship between the passive earth pressure and the retained height of the soil at different values of the interface friction angle δ as well as soil friction angle φ. It is apparent that the

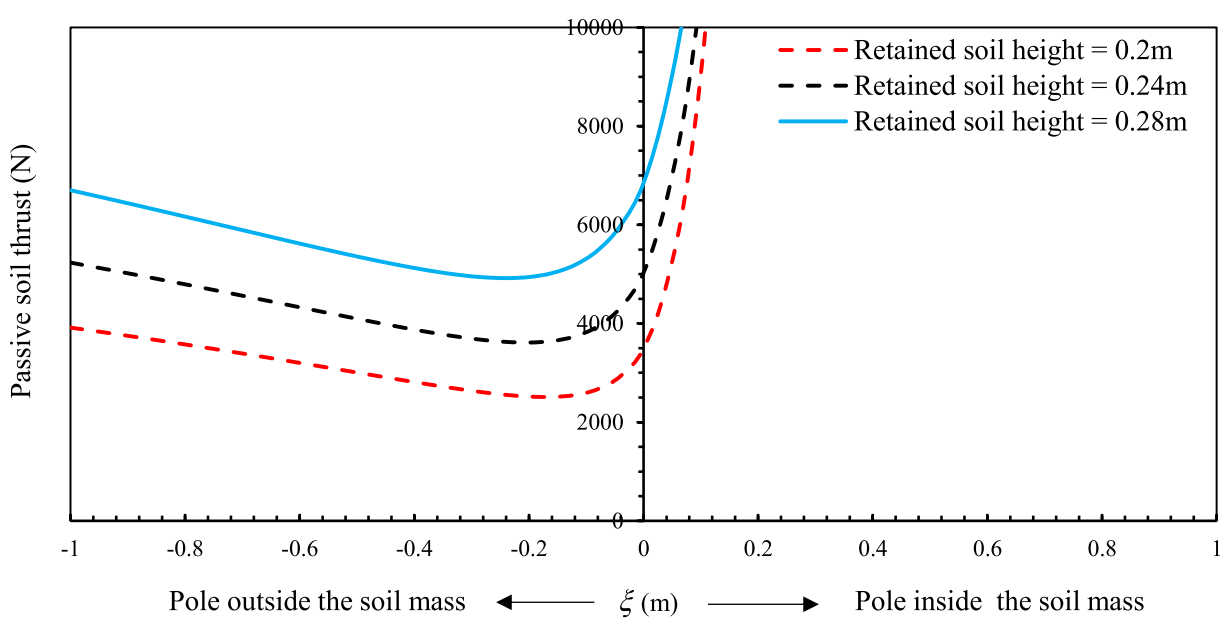

Fig. 11. Passive force vs the radial distance (ξ).

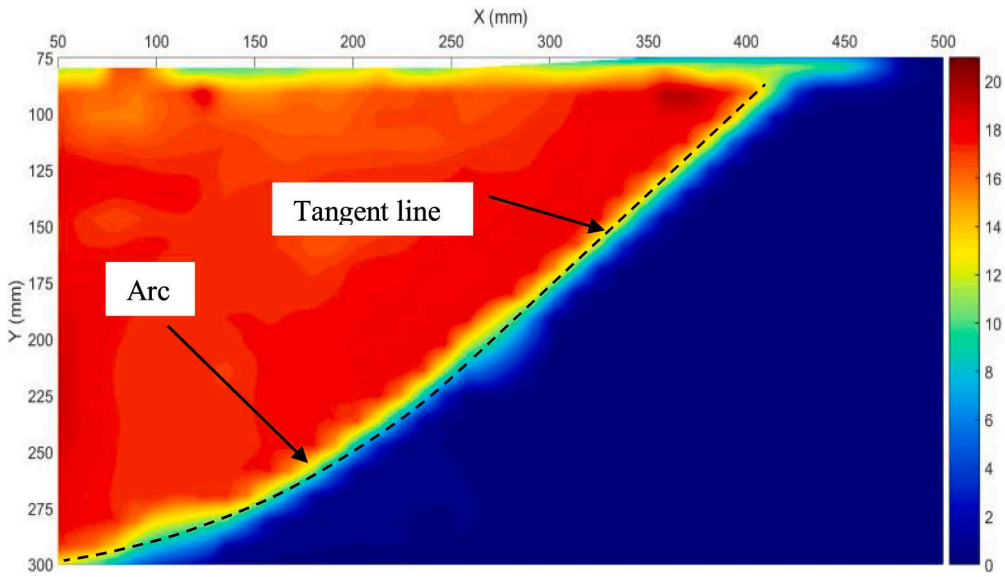

Fig. 12a. Total displacement contour map based on PIV analysis (Sigdel et al., 2021).

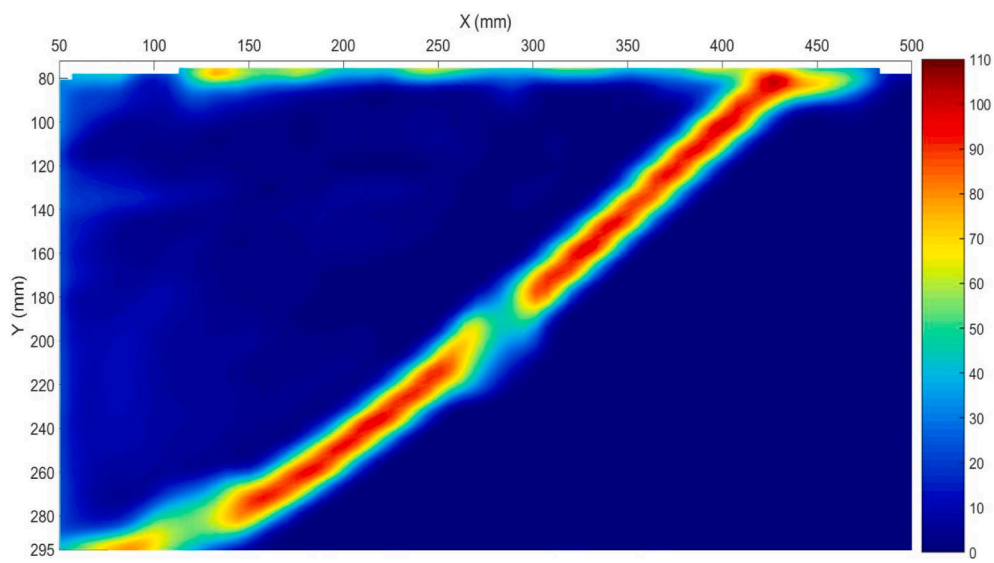

Fig. 12b. Shear band plot based on PIV analysis (Sigdel et al., 2021).

Table 3
Properties of sand used by Fang et al. (2002).

Soil Sample	Unit weight kN/m^3	Soil friction angle°		Interface friction δ°
		Peak	Critical	
Loose	15.7	33	31.5	9.8
Medium dense	16.3	38	31.5	12.6
Dense	16.8	42	31.5	14

Table 4
Comparison between the calculated results and the data reported by Fang et al. (2002).

Soil Sample	Measured force (N)	Calculated force (N) (log-spiral, this paper)	Error %
Loose	6300	6826	8.3
Medium dense	10,125	11,044	9.07
Dense	15,600	15,011	−3.7

passive force acting on the wall relates proportionally to the wall-soil interface friction angle at a given soil friction angleφ. According to Fig. 13($\varphi = 25^{\circ}$), when $\delta = \varphi$ the passive soil force is 77% higher than that obtained with $\delta = 0$ for a 4 m retained soil height. This justifies the low passive forces calculated using Rankine method, which assumes a smooth interface between the backfill and the wall. However, the influence of the interface friction becomes more significant at a higher friction angleφ. According to Fig. 13 ($\varphi = 30^{\circ}$) and ($\phi = 35^{\circ}$), the passive earth pressure at $\delta = \varphi$ are 119% and 185% greater than those calculated at $\delta = 0$, respectively.

Fig. 14 illustrates the effect of the ratio δ/ϕ on the passive earth force P_p. At a given magnitude of soil friction angleφ, the curvature of the log-spiral arc is governed by the interface friction angle δ. While the angles $\alpha_1$and $\alpha_2$are independent of δ, the passive Rankine zone enlarges when the ratio δ/ϕ increases. This, in turn, results in a larger passive soil wedge, $abg'e'$, and consequently a greater passive earth force. The increase in the size of the spiral arc emerges from a reduction in the radial distanceξ. Hence the Pole moves closer to the soil mass so that r_o decreases while the area abg increases.

Table 5
Comparison between the values of k_p calculated using the log-spiral solution and those calculated using different methods (Shiau et al., 2008).

$\phi = 40^\circ, \beta = 0, \omega = 0$

δ/φ	Coulomb theory	Caquot and Kerisel (1948)	Log spiral method (Duncan and Mokwa, 2001)	Sokolovski (1960)	Upper bound Chen (1975)	Lower bound -Upper bound Shiau et al. (2008)	Log-spiral solution (this paper)
0	4.60	4.59	4.60	4.60	4.60	4.60–4.61	4.61
1/3	8.15	8.13	8.17	_	7.73	6.87–7.79	7.58
1/2	11.77	10.36	10.50	9.69	10.08	8.79–10.03	9.74
2/3	18.72	13.10	13.08	_	13.09	11.30–12.87	12.24
1	92.72	17.50	17.50	18.20	20.91	18.64–20.10	18.86

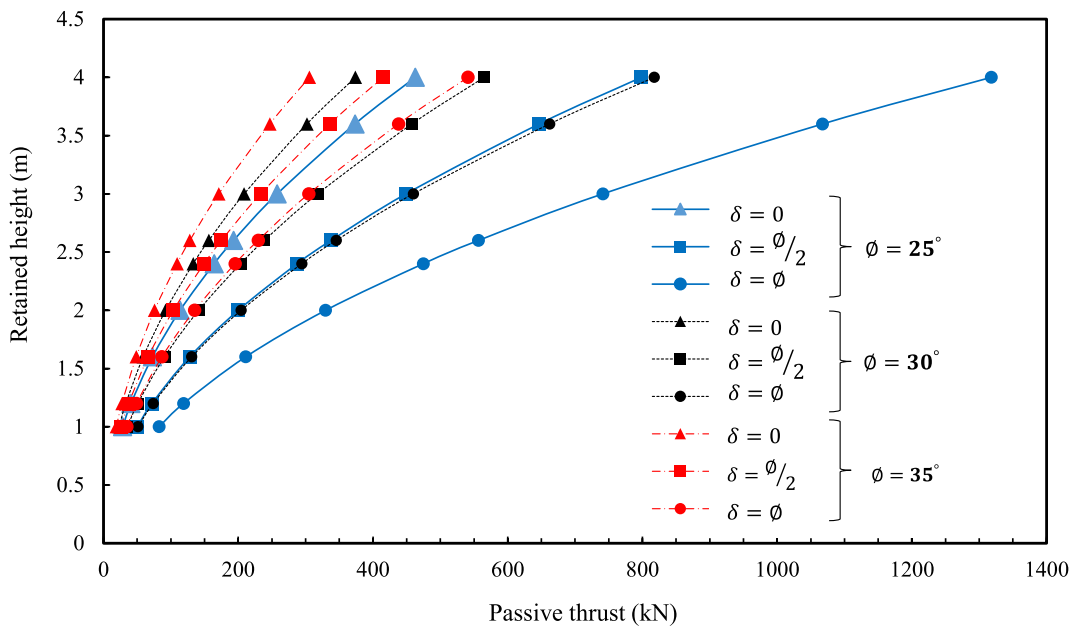

Fig. 13. The effect of the interface friction on the passive earth pressure.

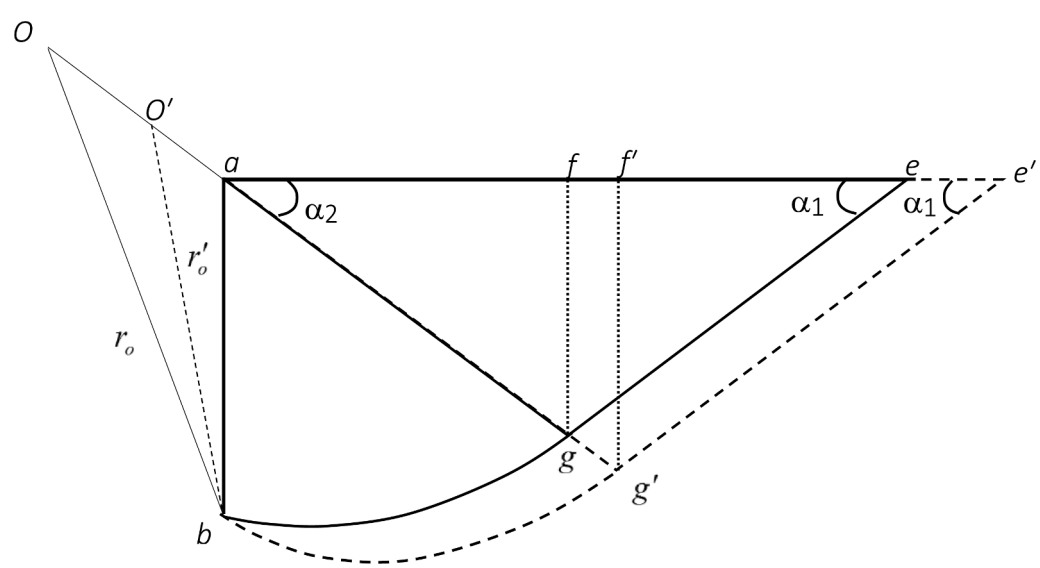

Fig. 14. The effect of the interface friction on the size of the passive wedge.

This fact leads to another conclusion about the interrelation between the ratio δ/ϕ and the radial distance ξ. This relation was examined by plotting the $P_p - \xi$ diagram over a range of δ/ϕ ratios. According to Fig. 15, in regular conditions (when the ratio δ/ϕ is between zero and 1), the Pole is generally located outside the soil mass. However, the location of the Pole moves toward the soil mass as the ratio δ/ϕ increases. At

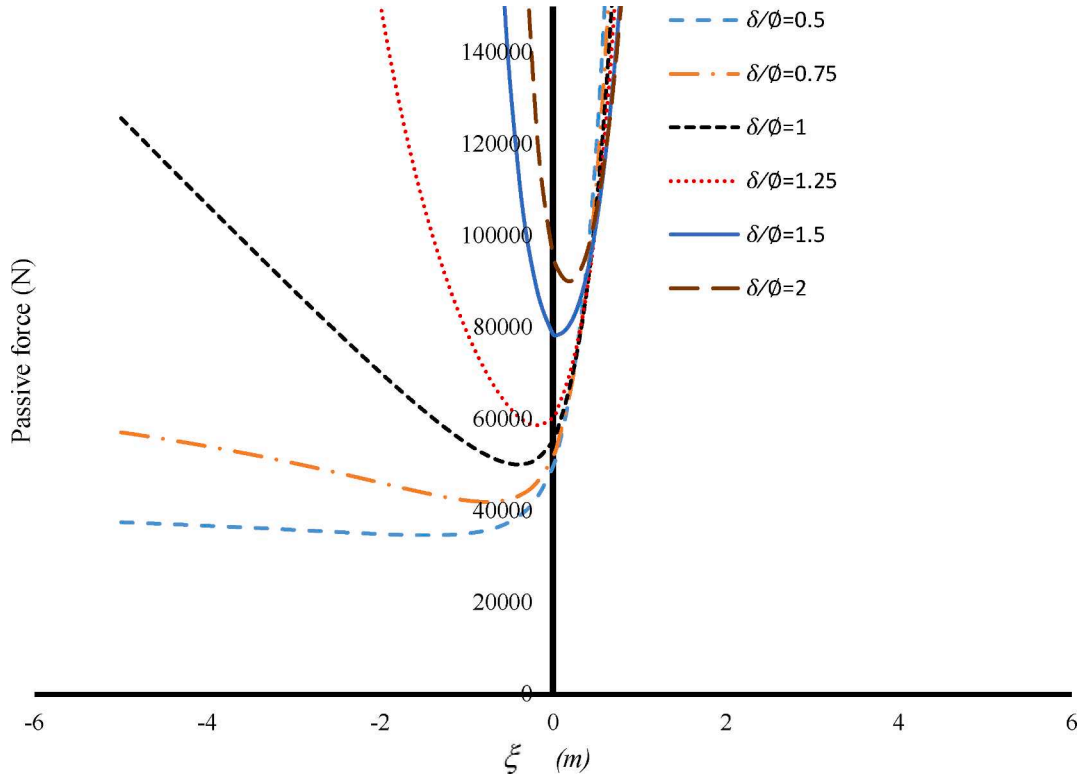

Fig. 15. The relationship between the interface friction and the location of Spiral Pole.

some extreme cases where $\delta/\phi = 2$, the Pole becomes clearly inside the soil mass. This phenomenon is associated with a dramatic increase in the magnitude of the passive soil force P_p.

4.3. Effects of backfill slope angle

The influence of the backfill slope β on the response of the retained soil to a passive wall movement has been reported by a number of studies. Shiau et al. (2008) investigated the effects of the backfill slope on the passive earth pressure coefficient using a finite element analysis. Fang et al. (1997) conducted experiments on a sloped backfill to study the effect of the backfill slope on the coefficients of lateral earth pressure $k_p$and k_a. Those researchers concluded that, both the magnitude of the horizontal passive earth pressure coefficient, k_p, and the wall movement needed for the backfill to reach a passive state increase when the backfill inclination angle β increases.

The data reported by Fang et al. (1997), for a normalised horizontal wall displacement (S/H) equals to 0.3, and Shiau et al. (2008) were used to quantitatively validate the log-spiral solution at different backfill slope angles. Fig. 16 shows that the results of the log-spiral solution fits perfectly between the upper and lower bound results reported by Shiau et al. (2008) but underestimate the experimental data of Fang et al. (1997). This disparity could be attributed to the frictional and the boundary reaction effects in the experimental tests of Fang et al. (1997), which may overestimate the actual values of k_p. The results obtained from the log spiral solution have also shown qualitative agreement with the findings of Fang et al. (1997) and Shiau et al. (2008) in relation to the effects of the backfill slope angle β. Fig. 17 shows that the magnitude of the passive earth force P_p increases with the magnitude of β at a given δ/ϕ ratio. The effect of β is greater for higher internal friction angle.

4.4. Effects of Wall Inclination

Shiau et al. (2008) reported a relationship between the internal friction angle and the passive earth pressure coefficient, k_p, at a different

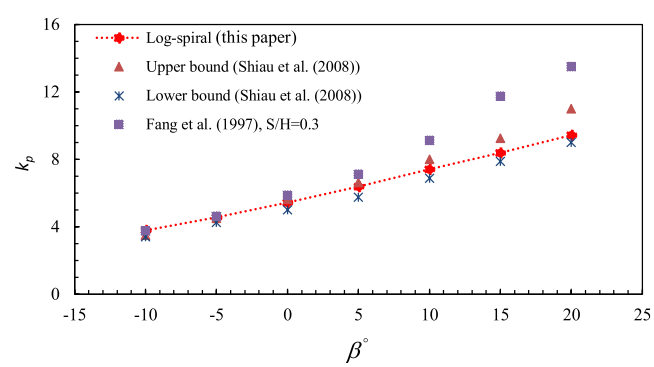

Fig. 16. Comparison with experimental results (after Fang et al. (1997) and Shiau et al. (2008)).

wall angles, ω. The log-spiral solution was compared with the corresponding results reported by Shiau et al. (2008). Fig. 18 shows that the results of the proposed log-spiral solution agree well with those reported by Shiau et al. (2008) with an error margin not exceeding 10%.

The effects of wall inclination have also been investigated at different interface friction angles (δ) as well as backfill slopes. It has been concluded that the inclination angle, ω, contributes to reduce the passive earth force P_p as well as the passive earth pressure coefficient, k_p. Such influence, however, is insignificant when the wall is frictionless ($\delta = 0$). The effects of the angle ω becomes prominent when the ratio δ/ϕ increases. Fig. 19 shows that when $\delta/\phi = 1$, the passive earth force decreases by approximately 35% as the angle ω increases from zero to 20°. This behaviour was observed to be independent of the backfill slope angle β.

5. Calculation of passive pressure coefficient k_p

The solution developed in this study is applied to calculate the co-

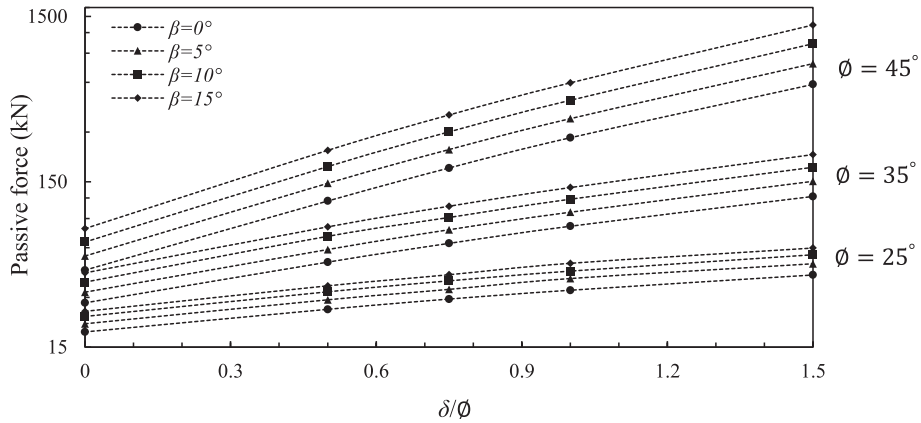

Fig. 17. The effect of the backfill slope on the passive earth pressure.

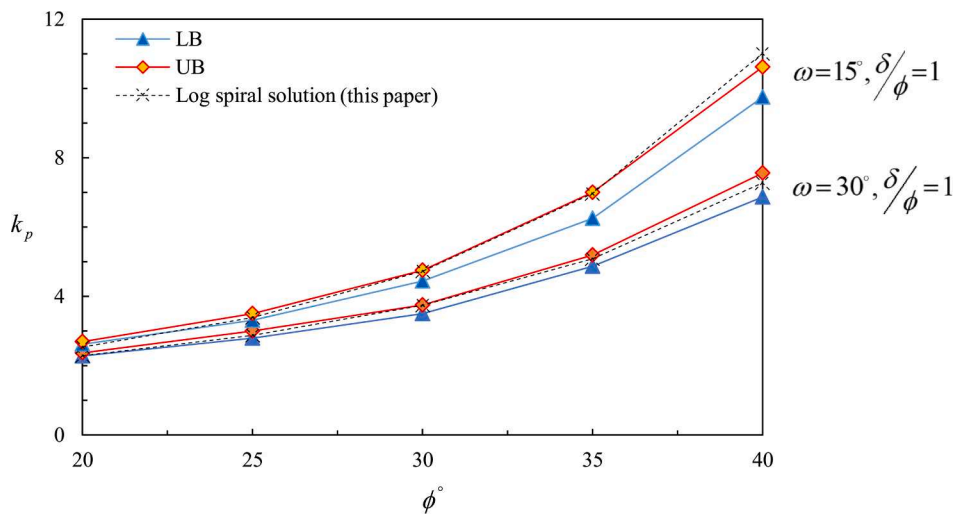

Fig. 18. Validation of the log-spiral solution against results from Shiau et al. (2008).

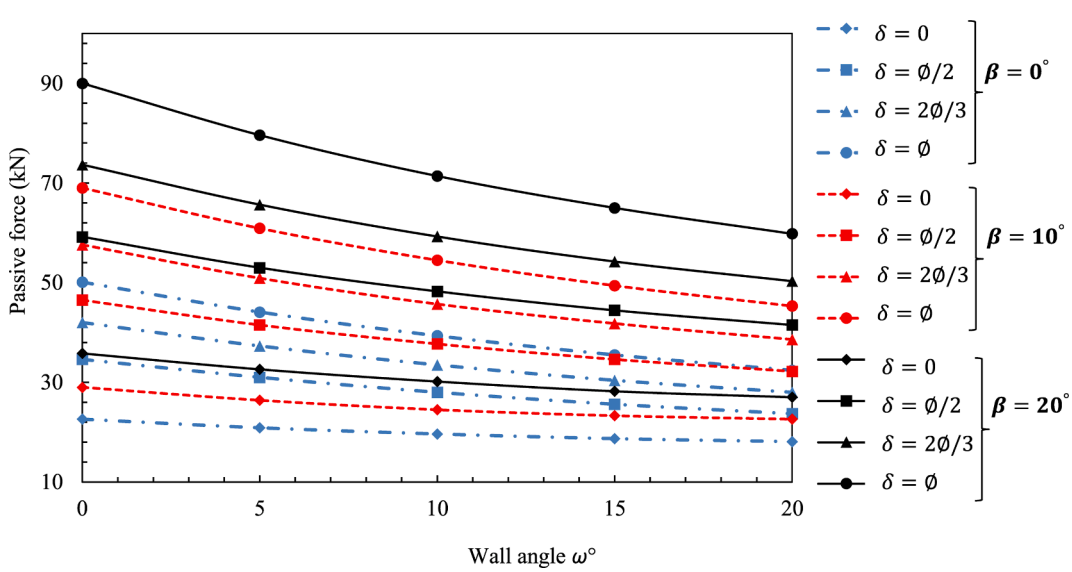

Fig. 19. The effect of the wall inclination angle on the passive earth pressure.

efficient of passive earth pressure k_p. Tables 6–9 provide a ready-made data sheets to identify the magnitude of k_p with respect to different wall, slope and soil conditions.

6. Conclusions

A comprehensive and rigorous generalised solution was formulated

Table 6
Coefficient of passive pressure $k_p(\varphi = 40^\circ)$.

$\frac{\delta}{\varphi}$	$\varphi = 40^\circ$															
	$\omega = 0^\circ$				$\omega = 5^\circ$				$\omega = 10^\circ$				$\omega = 15^\circ$			
	$\beta = 0^\circ$	$\beta = 5^\circ$	$\beta = 10^\circ$	$\beta = 15^\circ$	$\beta = 0^\circ$	$\beta = 5^\circ$	$\beta = 10^\circ$	$\beta = 15^\circ$	$\beta = 0^\circ$	$\beta = 5^\circ$	$\beta = 10^\circ$	$\beta = 15^\circ$	$\beta = 0^\circ$	$\beta = 5^\circ$	$\beta = 10^\circ$	$\beta = 15^\circ$
0	4.61	5.47	6.45	7.55	4.1	4.8	5.61	6.54	3.7	4.32	5	5.78	3.46	4.03	4.65	5.31
1/3	7.58	9.27	11.2	13.5	6.44	7.84	9.5	11.5	5.56	6.76	8.17	9.72	4.86	5.92	7.13	8.5
1/2	9.74	12.2	14.6	17.4	8.14	10.0	12.3	14.7	6.97	8.56	10.5	12.5	6.03	7.42	9.05	10.9
2/3	12.2	15.2	18.6	22.6	10.3	12.7	15.5	18.8	8.67	10.8	13.2	16	7.46	9.3	11.5	13.8
1	18.8	23.7	29.7	35.9	15.6	19.5	24.2	29.6	13	16.3	20.3	25.8	11	13.9	17.2	21

Table 7
Coefficient of passive pressure $k_p(\varphi = 35^\circ)$.

$\frac{\delta}{\varphi}$	$\varphi = 35^\circ$															
	$\omega = 0^\circ$				$\omega = 5^\circ$				$\omega = 10^\circ$				$\omega = 15^\circ$			
	$\beta = 0^\circ$	$\beta = 5^\circ$	$\beta = 10^\circ$	$\beta = 15^\circ$	$\beta = 0^\circ$	$\beta = 5^\circ$	$\beta = 10^\circ$	$\beta = 15^\circ$	$\beta = 0^\circ$	$\beta = 5^\circ$	$\beta = 10^\circ$	$\beta = 15^\circ$	$\beta = 0^\circ$	$\beta = 5^\circ$	$\beta = 10^\circ$	$\beta = 15^\circ$
0	3.7	4.29	4.95	5.63	3.34	3.86	4.42	5.02	3.1	3.56	4.06	4.58	2.9	3.36	3.83	4.3
1/3	5.42	6.44	7.54	8.7	4.75	5.64	6.6	7.64	4.21	5	5.86	6.78	3.79	4.5	5.29	6.12
1/2	6.54	7.85	9.36	10.9	5.67	6.78	8.01	9.37	5	6	7.06	8.2	4.46	5.34	6.31	7.36
2/3	7.8	9.36	11.1	13	6.76	8.12	9.62	11.2	5.92	7.16	8.48	9.9	5.22	6.3	7.52	8.85
1	10.8	13.1	15.7	18.5	9.2	11.2	13.4	15.8	8.26	9.69	11.6	13.7	7.13	8.75	10.6	12.1

Table 8
Coefficient of passive pressure $k_p(\varphi = 30^\circ)$.

$\frac{\delta}{\varphi}$	$\varphi = 30^\circ$															
	$\omega = 0^\circ$				$\omega = 5^\circ$				$\omega = 10^\circ$				$\omega = 15^\circ$			
	$\beta = 0^\circ$	$\beta = 5^\circ$	$\beta = 10^\circ$	$\beta = 15^\circ$	$\beta = 0^\circ$	$\beta = 5^\circ$	$\beta = 10^\circ$	$\beta = 15^\circ$	$\beta = 0^\circ$	$\beta = 5^\circ$	$\beta = 10^\circ$	$\beta = 15^\circ$	$\beta = 0^\circ$	$\beta = 5^\circ$	$\beta = 10^\circ$	$\beta = 15^\circ$
0	3	3.43	3.86	4.29	2.76	3.15	3.54	3.92	2.59	2.96	3.33	3.68	2.47	2.83	3.19	3.53
1/3	4.02	4.66	5.33	5.97	3.61	4.2	4.79	5.37	3.28	3.81	4.37	4.9	3.02	3.52	4.04	4.54
1/2	4.62	5.4	6.2	7	4.13	4.81	5.53	6.23	3.73	4.36	5.02	5.66	3.41	4	4.61	5.21
2/3	5.27	6.17	7.11	8.03	4.71	5.52	6.36	7.18	4.22	4.69	5.75	6.52	3.83	4.51	5.23	5.93
1	6.68	7.9	9.2	10.4	5.88	6.97	8.1	9.22	5.34	6.35	7.41	8.47	4.78	5.7	6.65	7.6

Table 9
Coefficient of passive pressure $k_p(\varphi = 25^\circ)$.

$\frac{\delta}{\varphi}$	$\varphi = 25^\circ$															
	$\omega = 0^\circ$				$\omega = 5^\circ$				$\omega = 10^\circ$				$\omega = 15^\circ$			
	$\beta = 0^\circ$	$\beta = 5^\circ$	$\beta = 10^\circ$	$\beta = 15^\circ$	$\beta = 0^\circ$	$\beta = 5^\circ$	$\beta = 10^\circ$	$\beta = 15^\circ$	$\beta = 0^\circ$	$\beta = 5^\circ$	$\beta = 10^\circ$	$\beta = 15^\circ$	$\beta = 0^\circ$	$\beta = 5^\circ$	$\beta = 10^\circ$	$\beta = 15^\circ$
0	2.47	2.77	3.06	3.3	2.31	2.6	2.87	3.11	2.19	2.48	2.75	3	2.12	2.41	2.68	2.91
1/3	3.06	3.48	3.88	4.2	2.81	3.2	3.57	3.88	2.61	3	3.34	3.63	2.46	2.81	3.15	3.44
1/2	3.4	3.86	4.32	4.7	3.1	3.55	3.97	4.33	2.87	3.29	3.7	4.02	2.68	3.08	3.47	3.8
2/3	3.73	4.28	4.81	5.24	3.41	3.91	4.4	4.8	3.13	3.6	4.05	4.43	2.92	3.37	3.8	4.16
1	4.52	5.21	5.88	6.43	4.05	4.68	5.29	5.79	3.7	4.27	4.83	5.3	3.4	3.95	4.48	4.91

based on the Terzaghi method to estimate the lateral passive pressure of the soil exerted on a retaining structure. The developed method can be used to calculate the passive force under different values of soil friction angle, soil-wall interface friction, backfill slope and wall inclination angles. A series of experiments were carried out to validate the developed method. Also, the developed equations were validated against measured data reported in the literature. Based on the results following conclusions can be reached,

The log-spiral solution developed herein is an accurate method to calculate the lateral earth pressure. The results calculated using the proposed method agree with the measured data better than those calculated using Rankine or Coulomb methods.

The passive earth pressure depends heavily on the internal friction angle of the soil and the interface friction between the backfill and the retaining wall.

The location of the Pole of the log-spiral arc is often located outside the soil mass. However, in cases where the interface friction is too high, the Pole may be located inside the soil mass.

The failure plane derived using the images captured during experiments confirms that it is not a straight linear failure surface but a curved surface closer to the toe of the wall and a straight line towards the retained soil surface, which resembles the failure surface assumed for the log-spiral solution presented in this paper.

The log-spiral solution can be easily programmed using MATLAB or even using an EXCEL spreadsheet, which provides an efficient and easy tool for engineers to estimate the passive earth pressure accurately.

CRediT authorship contribution statement

A.S. Alqarawi: Conceptualization, Methodology, Formal analysis, Validation, Visualization, Writing - original draft. **C.J. Leo:** Conceptualization, Methodology, Writing - review & editing. **D.S. Liyanapathirana:** Conceptualization, Writing - review & editing. **L. Sigdel:** Validation. **M. Lu:** Methodology, Validation. **P. Hu:** Validation, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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